Patrick Thangarajah

Simulation & FEA | Testing & Validation | Mechanical Design | Prototyping

403-585-1903 | petthang@uwaterloo.ca | www.linkedin.com/in/patrickthangarajah/ | Eligible to work in the U.S. under TN status

Summary

Mechanical Engineering MASc graduate with experience in finite element modeling, model validation, impact mechanics, and dynamic simulation.

Skills

Programming and Data Analysis MATLAB, Python, JMP, DOE, Microsoft Excel, C/C++

Model Validation and Testing FE model validation, physical test correlation, ISO 18571, experimental data analysis

Technical Communication Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs

Education

Master of Applied Science in Mechanical and Mechatronics Engineering *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Jul 2026*

- Graduate coursework average: 90%
- Research in impact mechanics, finite element analysis, and model validation

Bachelor of Applied Science in Mechanical Engineering, Co-op Program *Vancouver, BC, Canada*

University of British Columbia *Sep 2018 - May 2023*

- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Jul 2026*

- Designed and analyzed a 104-simulation FE parametric study of seat belt anchor locations using JMP DOE, least-squares regression, and ANOVA; identified a rearward and inboard position with 16% lower peak chest deflection than the study average
- Validated a THOR-5F ATD finite element model against physical sled testing in two bench configurations in LS-DYNA, achieving an average ISO 18571 score of 0.71 across 10 measurement channels
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations

Summer Research Student *Vancouver, BC, Canada (Remote)*

Centre for Heart Lung Innovation *May 2022 - Aug 2022*

- Built a Jupyter Notebook transfer-learning script to diagnose diabetic kidney disease using an emerging imaging platform
- Improved diagnostic accuracy by 10% through model optimization, training refinement, and comparison with existing tissue-slide images

Teaching Assistant - MTE 120: Circuits *Waterloo, ON, Canada*

University of Waterloo *May 2024 - Aug 2024*

- Supported laboratory instruction for a 100+ student circuits course, troubleshooting hardware and providing feedback on lab work and exams

Mechanical Engineering Intern *Airdrie, AB, Canada*

GN Corporation *Jan 2022 - Apr 2022*

- Reduced fiberglass cutting waste by 50% through process analysis, and proposed a material handling workflow projected to halve pick-and-place time
- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes

R&D Engineer (Part-time) *Remote*

Echoscope *May 2023 - Sep 2023*

- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functionality testing; all five were approved and used by the client

Projects

Simulation of Cervical Fractures in American Football (ME 720 Course Project) *Waterloo, ON, Canada*

University of Waterloo *Sep 2023 - Dec 2023*

- Simulated head-first helmet impacts across three velocities (3.4–7.0 m/s) and three neck angles at the lowest speed condition using Hybrid III and GHBMC human body models in LS-DYNA, then evaluated fracture force thresholds and injury location against peer-reviewed literature
- Discovered that predicted peak neck forces (2.26–2.34 kN) fell within the range reported across two cadaveric studies, and that the Neck Injury Criterion exceeded its injury-risk threshold in only 1 of 5 configurations despite fracture occurring in all 5

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- Watson, B., Seeburrun, T., **Thangarajah, P.**, Tang, K., Tylko, S., & Cronin, D. S. (2026). *Experimental and numerical investigation of belt loading for small-stature female occupants in rear seat frontal crash*. Proceedings of the 28th International Technical Conference on the Enhanced Safety of Vehicles (ESV), Toronto, ON, Canada, May 12–15, 2026.
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.